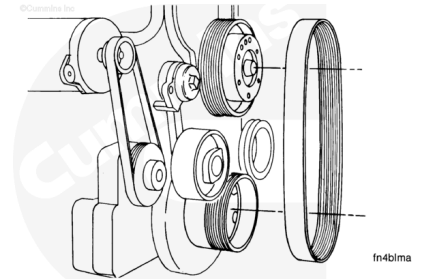


Trainings ▾

Preparatory Steps

Remove the fan belt. Refer to Procedure 008-002 in Section 8.
(</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>)



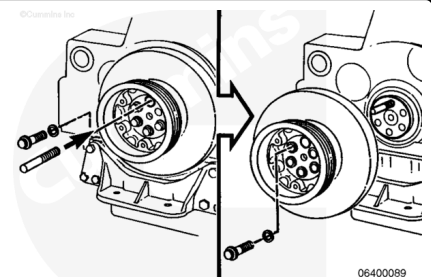
Remove

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

⚠ CAUTION ⚠

Do not pry or hammer on the vibration damper. Damage to the vibration damper can result.



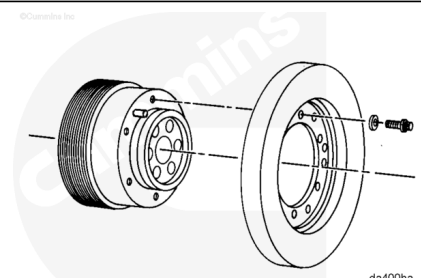
Remove one capscrew and install a $\frac{3}{4}$ -16x5 inch guide stud.

Remove the remaining capscrews from the vibration damper.

Remove the guide stud.

Note : Engines that do not have a belt driven fan hub have an adapter instead of a pulley.

Remove the pulley from the vibration damper.



Inspect for Reuse

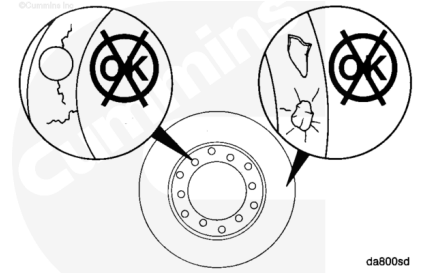
Note : Do not repair or balance a viscous vibration damper in the service field.

Vibration dampers have limited service life. The damper **must** be replaced after 576,000 km [360,000 mi], or 15,000 hours of service.

Clean and inspect the damper for reuse. Inspect for cracks on the mounting flange.

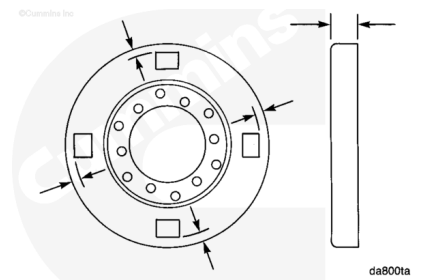
Inspect for dents or bulges on the housing.

If the vibration damper is damaged, it **must** be replaced.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



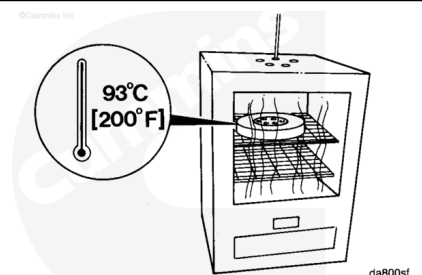
Use a paint solvent and fine crocus cloth to remove the paint from the front and back of the housing at the four areas shown in the illustration.

Measure no less than 3 mm [1/8 in] from the outside diameter to make sure that the measurement is on a flat surface.

Measure the thickness at four places as shown in the illustration. If the measurements vary more than 0.25 mm [0.010 in], or if the thickness exceeds 42.24 mm [1.663 in], the vibration damper **must** be replaced.

⚠ WARNING ⚠

Improper practices, carelessness, or ignoring the warnings can cause burns, cuts, mutilation, asphyxiation, or other personal injury or death.



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts that have been heated.

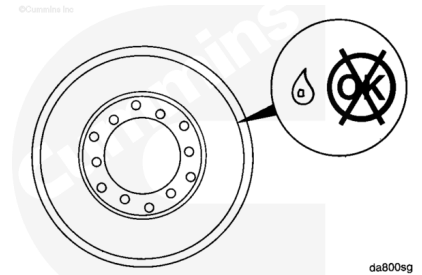
Use an oven with a temperature adjusted to 93°C [200°F].

Put the vibration damper in the oven with the rolled lip down.
Heat the damper for 2 hours.

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts that have been heated.

Remove the damper and check for leakage around the lip. If there is any leakage, the vibration damper **must** be replaced.



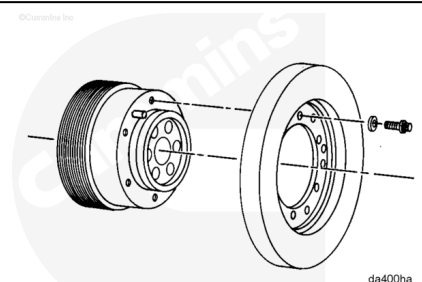
Install

Install the vibration damper on the crankshaft pulley or adapter, if equipped.

Install the six washers and capscrews.

Tighten the capscrews.

Torque Value: 140 n•m [105 ft-lb]



Install the guide stud in the crankshaft.

Install the vibration damper assembly. The pilot **must** be aligned with the crankshaft correctly.

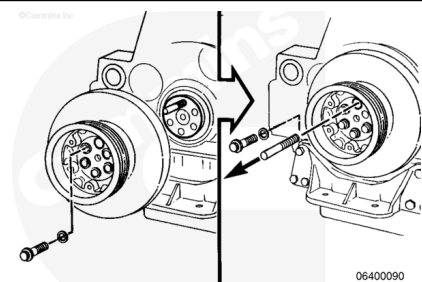
Install the washers and capscrews.

Tighten the capscrews.

Torque Value:

1. 230 n•m [170 ft-lb]

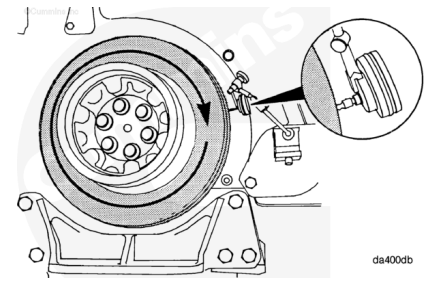
2. 445 n•m [330 ft-lb]



Eccentricity Check

Use a dial indicator and adjust it as shown to measure radial alignment of the vibration damper. Turn the crankshaft 360 degrees and record the total indicator readout.

Maximum Eccentricity		
mm		in
1.02	MAX	0.040

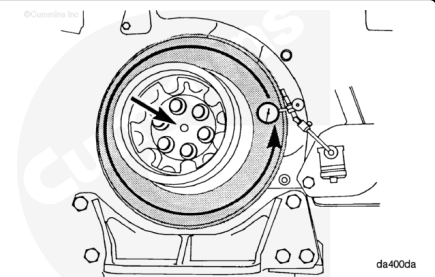


If the measurement is **not** within specifications, make sure the damper pilot is aligned properly.

Wobble Check

The crankshaft end clearance **must** be pushed or pulled in the same direction each time a measurement is taken.

Maximum Face Runout		
mm		in
1.02	MAX	0.040

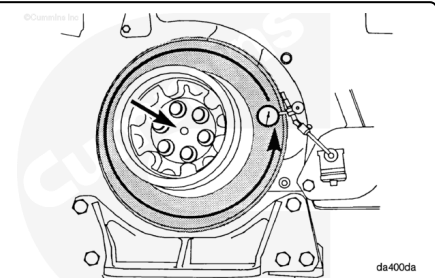


Note : When the dial indicator is positioned more than 13 mm [$\frac{1}{2}$ in] from the outside diameter, the specification is different.

To determine the specification:

- Measure the distance from the center of the crankshaft to the point the indicator tip touches the damper.
- Multiply the distance by 0.18 mm [0.007] when the damper is rubber; 0.15 mm [0.006 in] when the damper is viscous, to obtain the maximum total indicator readout.

If the measurement is not within specifications, check for foreign material between the crankshaft and the adapter (or pulley), and between the adapter (or pulley) and the vibration damper.



Finishing Steps

- Install the fan belt and related components. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>)

Last Modified: 09-Jul-2019
